

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> To solve, first calculate the time it would normally take to download the file without interruptions. Then, account for the interruptions and the time lost due to the restart. The file is 200 GB, and normally Carla can download at 2 GB/minute. So, without interruptions, it would take $200 / 2 = 100$ minutes. 40% of the way through is $200 * 0.4 = 80$ GB. At 2 GB/minute, this takes $80 / 2 = 40$ minutes. After this, a restart takes 20 minutes, and she has to restart the download from the beginning, so another 80 GB must be downloaded, which also takes 40 minutes. So, the total time is the sum of these times. </think><think> 100 minutes (initial download) - 40 minutes (first 40%) - 20 minutes (restart) + 40 minutes (second 40%) = 100 minutes. </think><answer>100</answer>

